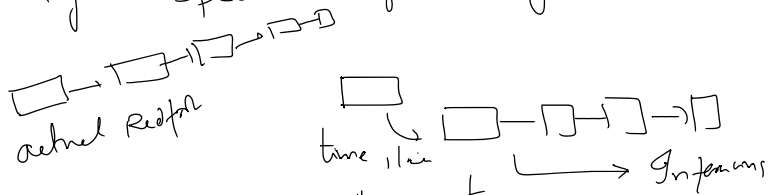


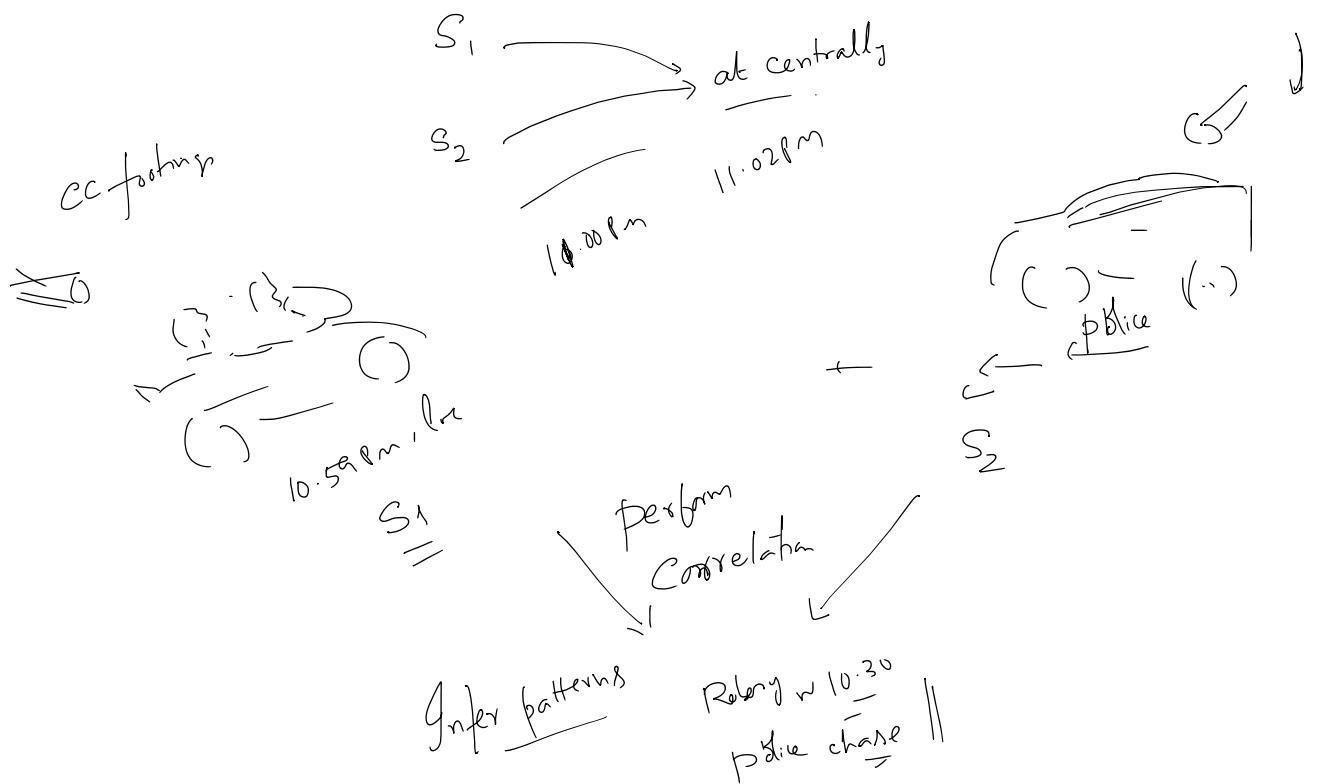
Agenda

- ① Need for Stream processing ✓
- ② Potential use case ✓
- ③ Sources of streaming Data ✓
- ④ Common interaction patterns (Data ingestion)
- ⑤ Complex Event processing - Special case of streaming ✓

Complex Event processing (CEP)



- Helps in discovering complex patterns - correlating into other events



"Complex Event" → Not that events are complex
 o/p Inference from set of events is complex

Cause and effect relationship

Stream processing vs Complex Event Processing

① stream processing is distributed and parallel computing

CEP is more of centralized activity

↳ Scalability is a challenge

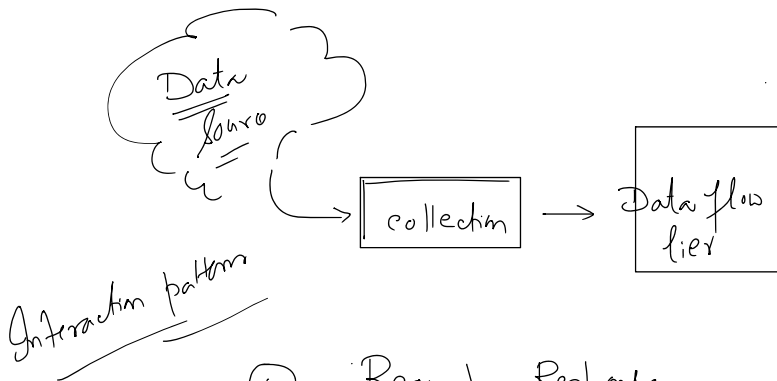
② stream processing ~~data~~ deals with Core jobs / tasks

CEP is querying (kind of format) due to nature of task [collective inferencing]

③ CEP Engines ~ low latency [Response time ~ ms]

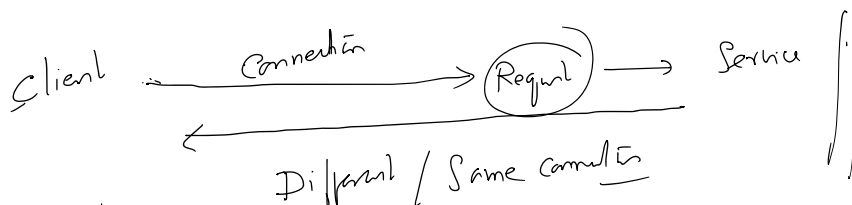
↳ load fluctuations could be a challenge

Generalized streaming architecture



① Request - Response

② Publisher - Subscriber
=



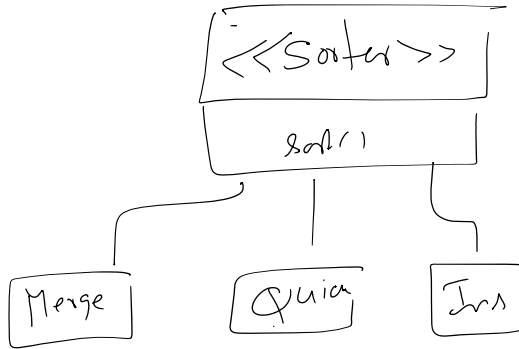
② publisher -> Subscriber

interface inheritance

② Publisher → Subscriber

Design patterns

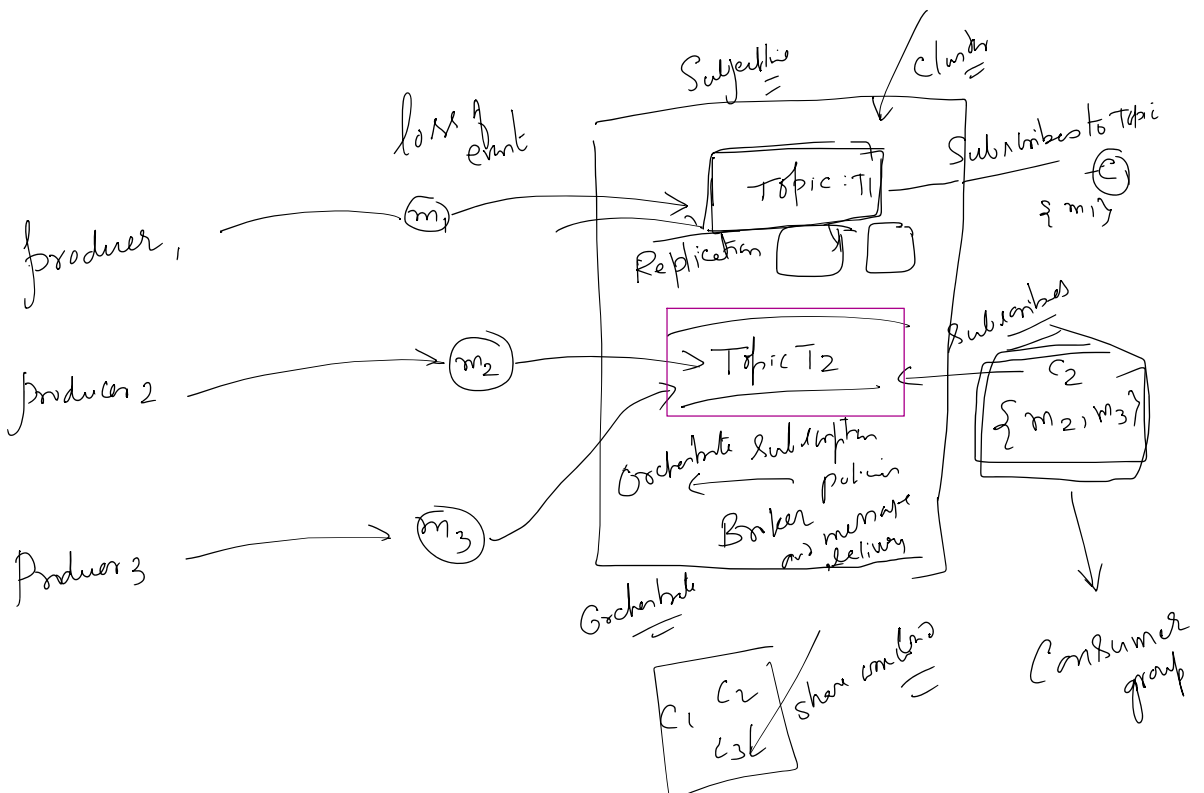
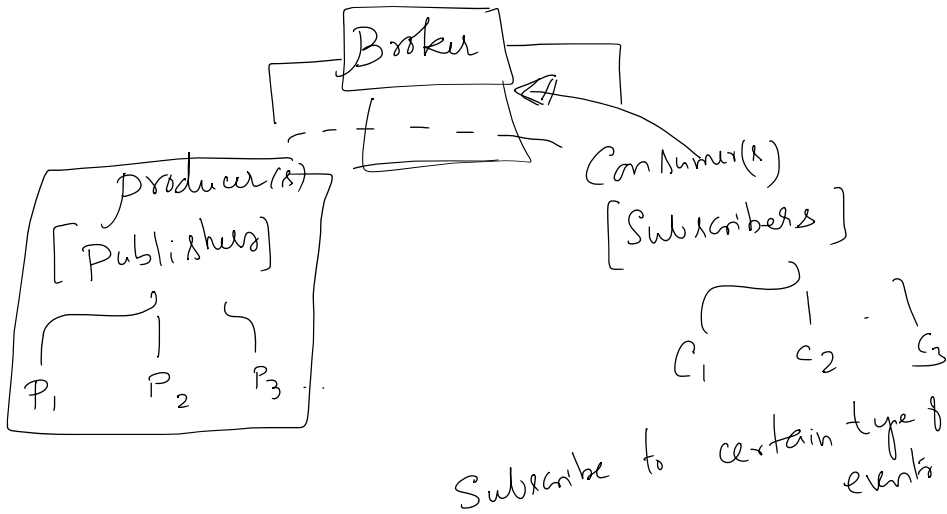
leverage interface inheritance
instead of class inheritance

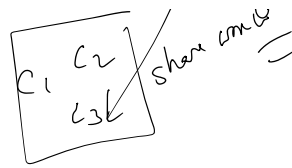


Sorter
s = new Insert()

obj

Complex
Diversified
Data
ingestion





Consumer group

Some of the topics very large (voluminous)

